

Secure Network Topology Design & Implementation

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1. Objective

The goal of this lab was to design and deploy a secure network architecture for a small office environment using Cisco Packet Tracer. The project focuses on isolating public servers from private company workstations and learning how to bring networking hardware online using real Cisco command line configurations.

2. Network Layout & Security Features

I built the network map from scratch and split the environment into two main zones to keep sensitive traffic completely isolated:

- **Internal Corporate LAN:**
On the left side, I set up a dedicated subnet for internal employee workstations (PC0 and PC1) connected to a central access switch. This entire zone handles private day to day operations and sits safely behind the router.
- **The DMZ (Demilitarized Zone):**
On the right side, I isolated the public facing production web server onto its own completely separate switch interface. This is a crucial security step because it ensures that if an attacker compromises the web server over the internet, they are trapped in the DMZ and cannot easily pivot straight into the employee network.
- **Core Setup:**
The central router handles all traffic between the two networks. Since Cisco router ports are turned off by default, I opened up the CLI terminal to configure them manually. I used commands like *interface* and *no shutdown* to turn on both GigabitEthernet0/0/0 and GigabitEthernet0/0/1, which successfully brought the offline connections online and turned the link lights green.

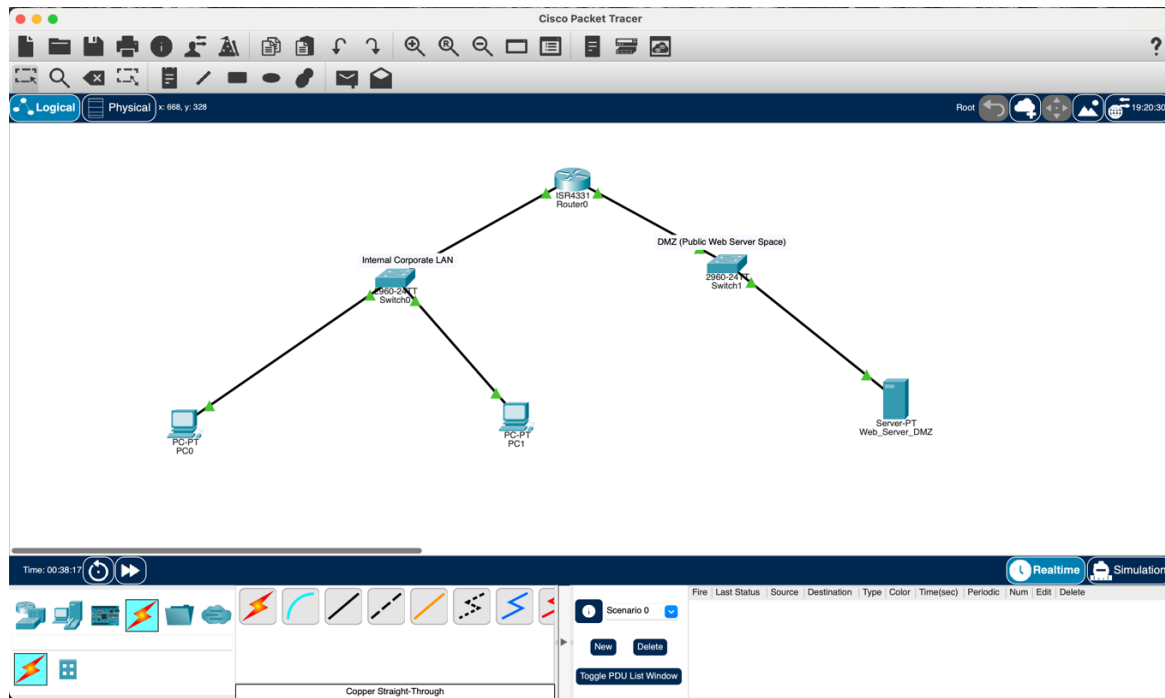


Figure 1: Final active network topology showing LAN and DMZ segment isolation.

3. Conclusion & Takeaways

Building this lab highlights how critical network design is to overall security defenses. By splitting the network into isolated zones and managing hardware interfaces directly through the command line, the business reduces its direct exposure to widespread network sniffing and lateral movement attacks.